Explainability of Large Language Models for Classification Tasks

Group 1, Project TA: Faeze

Dongxuan Li[1] Zemeng Wang[2] Chaochen Zhang[3]

[1]Department of Mechanical and Industrial Engineering

[2]Department of Electrical and Computer Engineering

[3]Department of Electrical and Computer Engineering

Applied Deep Learning – Winter 2026

UNIVERSITY OF TORONTO

Project Content and Team Contribution

Content:

Task: Joint text classification and natural language explanation generation.

Dataset: CoS-E (Common Sense Explanations) featuring human-written rationales.

Model: GPT-2 backbone with LoRA (Low-Rank Adaptation) for efficient tuning.

Goal: Achieve high predictive accuracy while maintaining semantic explanation quality.

Team Contribution:

Dongxuan: Model development, PEFT framework integration, and training pipeline.

Zemeng: CoS-E data pre-processing, prompt engineering, and CoT (Chain-of-Thought) implementation.

Chaochen: Experimental evaluation, BERTScore validation, and error analysis.

UNIVERSITY OF TORONTO

Example and Pipeline

CoS-E Example

question: "There are 10 apples on an apple tree. Three fall off. Now there are X apples." What is this an example of?
choices: [park, coloring book, garden center, math problem, gravity]
answer: math problem
abstractive_explanation: webmath is designed to help you solve

Pipeline:
Prompt Construction: Formatted as [Question + Choices].
Training Target: Joint generation of [Correct Answer + Explanation].
Loss Calculation: Prompt Masking applied to focus training only on output tokens.
Evaluation: Classification Accuracy for label prediction + BERTScore F1 for explanation similarity.

UNIVERSITY OF TORONTO

System Design and Method

Model and Training Setup:

- Architecture: GPT-2 base model with LoRA (PEFT) integration.
- Optimization: AdamW optimizer with Linear LR Scheduler and Gradient Clipping (1.0).
- LoRA Rank Ablation: Systematic testing of ranks $r \in \{4, 8, 16\}$.
- CoT Prompting: Evaluated "Let's think step by step" to test emergent reasoning.

Key Results:

Model	Prompt	LoRA Rank	Accuracy@200
GPT-2	Normal	-	0.16
GPT-2	CoT	-	0.06
GPT-2 + LoRA	Normal	8	**0.36**
GPT-2 + LoRA	CoT	8	0.35

UNIVERSITY OF TORONTO

Results and Insights

Explanation Similarity:
Before and after applying LoRA, BERTScore F1 remains around 0.8, suggesting explanation similarity is relatively stable.

Error Analysis:
Semantic and Commonsense: Misinterpreting related terms or basic associations.
Spatial and Bias: Struggles with location logic and favors high-frequency words.

Conclusion:
Efficiency: LoRA is highly effective for resource-constrained fine-tuning.
Limitation: A trade-off exists between accuracy and matching human explanation style.

UNIVERSITY OF TORONTO